

SSH - Secure Shell

System Hypothesis

- Synchronous communication channel
- **No pre-shared public keys (at first)**
- Mallory can read, modify and forge message send over the network

Goals

- When two participant exchange a message, the system should protect the confidentiality, integrity and perfect forward secrecy of the messages

Why not?



K_{SB}, K_{pB}

$\text{Sign}_{skB}(\text{"I'm Bob"}, K_{pB})$



K_{SM}, K_{pM}

$\text{Sign}_{skM}(\text{"I'm Bob"}, K_{pM})$